
Reliable Causal Estimation under Latent Markov Confounding

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Abstract

When a persistent latent regime drives both a treatment and an outcome, a standard remedy is to infer the regime with a hidden Markov model and condition on its posterior. We show that this soft conditioning leaves a bias with an exact closed form. The bias is governed by the *residual state variance* v left in the posterior and by two regime shifts: one in the treatment (Δ_T) and one in the outcome (Δ_g). It equals $\Delta_T \Delta_g v / (\Delta_T^2 v + \sigma^2)$, where σ^2 is the exogenous treatment variance. The bias is non-monotone in confounding strength, so the conditioning of the residualized treatment (the usual Gram-matrix $\lambda_{\min}$) is not a monotone predictor of reliability. The quantity v is observable for calibrated posteriors, with an error bounded by the calibration error, leaving Δ_g as the single unidentified parameter, which yields a one-parameter sensitivity analysis and a robustness value. In learned-HMM simulations with dependent data, linear and boosted nuisances, and two or three regimes, the law predicts the realized bias with a median relative error of about 5%. It also reproduces the bias of a difficulty-frontier benchmark, and it shows that a regime-aware coupled estimator is no better than regime fixed effects. On four Indian air-quality networks, the Delhi estimate is robust to residual regime confounding while the other cities are inconclusive.

1 INTRODUCTION

Observational time series are often shaped by a persistent hidden regime: a boundary-layer state in atmospheric data, a macro regime in finance, an operating mode in a plant. If the regime moves both the

treatment and the outcome, adjusting for observed covariates X_t does not identify the effect. A common remedy is to fit a hidden Markov model (HMM) to auxiliary proxies Z_t and then condition on the regime posterior $\hat{\gamma}_t$, either as a hard label or as a soft probability, inside a double machine learning (DML) estimator (Chernozhukov et al., 2018). The practical question is how far the result can be trusted when the posterior is imperfect.

Two quantities are commonly used to answer it: the uncertainty of the posterior (its entropy or L^1 error ε_γ) and the conditioning of the score Jacobian, whose smallest eigenvalue $\lambda_{\min}$ measures how much treatment variation is left after residualization. Heuristically, error then scales like $\varepsilon_\gamma / \lambda_{\min}$. We show that this heuristic is wrong in a specific, checkable way, and we replace it with an exact law.

Contributions. (1) *A residual-state bias law* (Thm. 1, Cor. 1). For soft conditioning on any posterior computed from the proxy sequence, with learned parameters and dependent data, $\hat{\theta} - \theta = \Delta_T \Delta_g v / (\Delta_T^2 v + \sigma^2) + O_P(N^{-1/2})$, with a matrix version for $K > 2$ regimes. (2) *Non-monotone difficulty*. The bias is hump-shaped in Δ_T with peak $|\Delta_g| \sqrt{v} / (2\sigma)$, whereas the residual treatment variance (the scalar analogue of $\lambda_{\min}(\hat{J})$) rises monotonically. Refining the conditioning information can only reduce the bias (Prop. 1); soft posteriors beat hard regime labels. (3) *Observability and sensitivity analysis* (Prop. 2, Sec. 4). v is estimated by the posterior Gini impurity, with error at most $\sqrt{CE} + CE$ in the posterior’s calibration error. Only Δ_g is unidentified, giving a robustness value that says how large a hidden outcome shift must be to explain an estimate away. (4) *Evidence with strong baselines* (Sec. 7). The law matches simulations to a median relative error of about 5%, explains the bias of our own difficulty benchmark, exposes an experimental pitfall (nuisances that omit the regime), and is applied to four Indian cities.

Relation to prior work. Mathematically the law is an omitted-variable-bias (OVB) identity with a partially observed confounder (Greenland, 1980; Cinelli and Hazlett, 2020; Chernozhukov et al., 2022); the

hump follows from the same algebra that underlies bias amplification (Pearl, 2011; Ding et al., 2017). We do not claim these ingredients as new. What we add is (i) the identification of the residual variance with the calibrated HMM posterior, valid for learned posteriors and dependent data under purged cross-fitting; (ii) the consequence that the Gram-matrix conditioning, a standard DML diagnostic, is not a monotone predictor of reliability here; and (iii) a calibration-based, nearly fully observable sensitivity analysis for latent Markov regimes. Posterior-based adjustment is known to fail without identification (Wang and Blei, 2019; D’Amour, 2019); our law quantifies how it fails. Latent-state and proximal approaches (Miao et al., 2018; Tchetgen Tchetgen et al., 2020) target identification, which we do not assume. HMMs follow Hamilton (1989); Cappé et al. (2005); dynamic DML is in Lewis and Syrgkanis (2020).

2 SETTING AND ESTIMATOR

Let $S_t \in \{0, \dots, K-1\}$ be a latent persistent regime with indicator vector $D_t = (\mathbb{1}\{S_t = k\})_{k=1}^{K-1}$. We observe a treatment T_t , an outcome Y_t , controls X_t , and proxies Z_t that depend on S_t . We posit

$$T_t = m(X_t) + a^\top D_t + V_t, \quad (1)$$

$$Y_t = \theta T_t + g(X_t) + b^\top D_t + U_t, \quad (2)$$

where $a \in \mathbb{R}^{K-1}$ and $b \in \mathbb{R}^{K-1}$ are the regime shifts of the treatment and the outcome (Δ_T and Δ_g when $K = 2$), θ is the causal effect, and (V_t, U_t) are exogenous noises (assumptions below). The regime is confounding when $a \neq 0$ and $b \neq 0$.

Soft conditioning. Let $\hat{\gamma}_t = \hat{\gamma}_t(Z_{1:N}) \in \Delta^{K-1}$ be any posterior computed from the proxies, for example the smoothing or filtering distribution of a Gaussian HMM whose parameters are learned by EM on the same sample. With features $F_t = (X_t, \hat{\gamma}_t)$ and purged block cross-fitting with an embargo of τ steps, let $\tilde{T}_t = T_t - \hat{m}_T(F_t)$ and $\tilde{Y}_t = Y_t - \hat{m}_Y(F_t)$ be cross-fitted residuals and

$$\hat{\theta} = \sum_t \tilde{T}_t \tilde{Y}_t / \sum_t \tilde{T}_t^2. \quad (3)$$

Hard assignment uses $\mathbb{K}\{\arg \max \hat{\gamma}_t\}$ instead of $\hat{\gamma}_t$. The coupled, regime-specific estimator of our earlier work (OR-DML) is analysed in Appendix E; it does not improve on these baselines.

3 THE RESIDUAL-STATE BIAS LAW

Let $c_t = \mathbb{E}[D_t | F_t]$, $\tilde{D}_t = D_t - c_t$, $\Sigma_N = N^{-1} \sum_t \tilde{D}_t \tilde{D}_t^\top$ (the residual state covariance), and $v = \Sigma_N$ when $K = 2$.

Assumption 1 (Exogenous noise). V_t is independent of (S, Z, X) at all times, $\mathbb{E}V_t = 0$, $\text{Var}V_t = \sigma^2 > 0$. $\mathbb{E}[U_t | S, Z, X, V] = 0$. Both noise processes have summable autocovariances and bounded $4+\delta$ moments.

Assumption 2 (Posterior as a local feature). The process is stationary and α -mixing. $\hat{\gamma}_t$ is a function of $Z_{1:N}$ that depends on Z_s for $|s - t| > \tau$ only through terms of size ε_τ in L^2 (filter stability), and estimated HMM parameters converge at rate $N^{-1/2}$.

Assumption 3 (Nuisance learners). The cross-fitted learners for $\mathbb{E}[T | F]$ and $\mathbb{E}[Y | F]$ have L^2 errors r_T, r_Y (set $r_h = r_Y + |\theta|r_T$) and the embargo τ exceeds the mixing time.

Theorem 1 (Residual-state bias law). *Under Assumptions 1–3,*

$$\hat{\theta} - \theta = \frac{a^\top \Sigma_N b}{a^\top \Sigma_N a + \sigma^2} + O_P(N^{-1/2} + r_T r_h + \varepsilon_\tau), \quad (4)$$

with constants depending on $\|a\|, \|b\|$ and σ^{-2} . In particular $\hat{\theta}$ is consistent iff $a^\top \Sigma_N b \rightarrow 0$, and the denominator equals $\text{Var}(\tilde{T})$.

The estimation error of the HMM enters *only through* v , because v is defined with respect to the realized posterior. The proof (Appendix A) writes the residuals as $\tilde{T} = a^\top \tilde{D} + V - \delta^T$ and $\tilde{Y} - \theta \tilde{T} = b^\top \tilde{D} + U - \delta^h$; all cross terms are centered mixing sums or nuisance products.

Intuition. After partialling out F_t , the treatment still contains the part of the regime that the posterior failed to explain, $a^\top \tilde{D}_t$, and so does the outcome, $b^\top \tilde{D}_t$. This shared component is a confounder that no longer has a proxy: it contributes covariance $a^\top \Sigma_N b$ to the numerator, while the denominator mixes it with the clean variation σ^2 . A strong treatment shift a makes the unexplained component dominate the treatment residual (so $\text{Var}(\tilde{T})$ is large), but then the estimator regresses $\tilde{Y}$ on a variable that is nearly a copy of $b^\top \tilde{D}$, and its slope tends to $\theta + b/a$; the bias is free of θ and shrinks like b/a as a grows. Weak shifts leave little bias to begin with.

Corollary 1 (Hump, sup bound, information monotonicity). *Let $K = 2$ and $B = \Delta_T \Delta_g v / (\Delta_T^2 v + \sigma^2)$ with $\lambda := \Delta_T^2 v + \sigma^2 = \text{Var}(\tilde{T})$. (i) $|B| \leq |\Delta_g| \sqrt{v} / (2\sigma)$ for all Δ_T , with equality at $|\Delta_T| = \sigma / \sqrt{v}$. (ii) $|B| = |\Delta_g| \sqrt{v} \sqrt{\lambda - \sigma^2} / \lambda$, which increases in λ on $[\sigma^2, 2\sigma^2]$ and decreases afterwards, while λ increases monotonically in $|\Delta_T|$. (iii) For fixed $(\Delta_T, \Delta_g, \sigma)$, $|B|$ is strictly increasing in v .*

Part (ii) is the correct replacement for the heuristic $\varepsilon_\gamma / \lambda_{\min}$: the numerator $\Delta_T \Delta_g v$ grows with the same confounding strength that inflates λ , so no function of $(\varepsilon_\gamma, \lambda)$ alone bounds the bias monotonically. The bound

(i) decays only as $\sqrt{v}$, so the worst case is $O(\sqrt{\varepsilon_\gamma})$ even though for fixed Δ_T the bias is linear in v .

Proposition 1 (Information monotonicity). *If $\mathcal{F} \subseteq \mathcal{F}'$ are two conditioning sets, then $\Sigma' \preceq \Sigma$, and for $K = 2$ $|B'| \leq |B|$. In particular, conditioning on the soft posterior never does worse than conditioning on its hard label, and adding informative controls or proxies cannot hurt.*

Proposition 2 (Observable plug-in). *Let $\hat{\Sigma} = N^{-1} \sum_t [\text{diag}(\hat{\gamma}_t) - \hat{\gamma}_t \hat{\gamma}_t^\top]_{1,1}$ (for $K = 2$, $\hat{v} = N^{-1} \sum_t \hat{\gamma}_t(1 - \hat{\gamma}_t)$, the posterior Gini impurity) and $CE = \mathbb{E} \|\hat{\gamma}_t - \mathbb{E}[D_t | F_t]\|^2$ the L^2 calibration error given F_t . For $K = 2$,*

$$|\mathbb{E}\hat{v} - \mathbb{E}v| \leq \sqrt{CE} + CE, \quad (5)$$

so $\hat{v} = v$ for a calibrated posterior and an over-confident one understates the bias. If m is linear and the posterior is calibrated, the OLS coefficient $\hat{a}$ of $\hat{\gamma}$ in the regression of T on $(1, X, \hat{\gamma})$ consistently estimates a , and $\sigma^2 = \text{Var}(\tilde{T}) - a^\top \Sigma a$.

Heterogeneous effects $\theta(S)$ add the terms $a(\theta_1 - \bar{\theta}) + \bar{m}(\theta_1 - \theta_0)$ to b (Appendix B), where $\bar{m}$ is the state-free mean treatment level.

4 SENSITIVITY ANALYSIS

By Thm. 1 and Prop. 2, everything in the bias except b is observable: $\hat{a}$ from the treatment regression, $\hat{\Sigma}$ from the posterior, and $\widehat{\text{Var}}(\tilde{T})$ from the residuals. For a hypothesized outcome shift b the bias-adjusted estimate is $\hat{\theta}_{\text{adj}}(b) = \hat{\theta} - \hat{a}^\top \hat{\Sigma} b / \widehat{\text{Var}}(\tilde{T})$. **Robustness value.** The smallest outcome shift that explains the point estimate away is

$$b^* = \frac{|\hat{\theta}| \widehat{\text{Var}}(\tilde{T})}{\|\hat{\Sigma} \hat{a}\|}, \quad (6)$$

reported in standard deviations of $\tilde{Y}$ and benchmarked against the observed regime shift of Y (the coefficient of $\hat{\gamma}$ in an outcome regression), which is not identified but is a natural yardstick. A large b^* means the estimate cannot be overturned by residual regime confounding through this channel; b^* is uninformative when $\hat{\theta} \approx 0$. A block bootstrap resamples contiguous blocks of rows and recomputes $(\hat{\theta}, \hat{v}, \hat{a}, \widehat{\text{Var}}(\tilde{T}))$.

5 A PRIORI BIAS FROM SENSOR PRIMITIVES

The law takes v as given. For a continuous latent confounder, v has a closed form in the sensor's information rate and the state's persistence, so the bias of posterior-conditioned DML can be predicted before (T, Y) are seen.

Algorithm 1 Residual-state sensitivity analysis

- 1: Fit an HMM to $Z_{1:N}$; recalibrate the posterior $\hat{\gamma}$ if its calibration error is large (Prop. 2).
 - 2: Cross-fit $\hat{m}_T, \hat{m}_Y$ on $(X, \hat{\gamma})$ with purged blocks; compute $\tilde{T}, \tilde{Y}, \hat{\theta}$.
 - 3: Compute $\hat{\Sigma}$ (posterior Gini), $\hat{a}$ (OLS of T on $(1, X, \hat{\gamma})$), $\widehat{\text{Var}}(\tilde{T})$.
 - 4: For a grid of outcome shifts b : report $\hat{\theta}_{\text{adj}}(b)$; report b^* in SD of $\tilde{Y}$ and its bootstrap interval.
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Proposition 3 (Wiener smoothing error). *Let u_t be a stationary unit-variance $AR(1)$ with coefficient ϕ , observed as $Z_t = hu_t + \varepsilon_t$ with $\varepsilon_t \sim N(0, R)$ i.i.d., and let $\iota = h^\top R^{-1}h$. Conditioning on the Kalman-smoothed mean leaves residual variance*

$$v = \frac{1 - \phi^2}{\sqrt{(1 + \phi^2 + \iota(1 - \phi^2))^2 - 4\phi^2}}, \quad (7)$$

and the bias is $\Delta_T \Delta_g v / (\Delta_T^2 v + \sigma^2)$ by Thm. 1 with u in place of S . For $0.02 \ll \iota \ll (1 - \phi^2)^{-1}$, $v \approx \frac{1}{2} \sqrt{(1 - \phi^2)/\iota}$, so the worst-case bias $|\Delta_g| \sqrt{v} / (2\sigma)$ decays only as $\iota^{-1/4}$; for $\iota \gg (1 - \phi^2)^{-1}$, $v \approx 1/\iota$ and it decays as $\iota^{-1/2}$.

The proof is the Wiener–Kolmogorov smoothing error $v = \int S_u / (1 + \iota S_u) d\omega / 2\pi$ with $S_u(\omega) = (1 - \phi^2) / |1 - \phi e^{-i\omega}|^2$, evaluated with $\int_0^{2\pi} d\omega / 2\pi (A - 2\phi \cos \omega)^{-1} = (A^2 - 4\phi^2)^{-1/2}$ (Anderson and Moore, 1979). In 18 simulated configurations ($\phi \in \{0.9, 0.97\}$, three sensor strengths, three Δ_T ; Table 1), the closed form matches the Kalman smoother's variance to 2×10^{-4} , and the predicted bias matches the DML bias with correlation 0.9987 and maximum error 0.021. For a persistent state, halving the worst-case bias therefore needs about $16\times$ the sensor information while $\iota \ll (1 - \phi^2)^{-1}$. Discretizing u into a K -state HMM posterior acts as a sieve: at $\phi = 0.97$, $\iota = 2.76$, $\Delta_T = 1$, the bias falls from 0.82 ($K=2$) to 0.52 ($K=3$) and 0.39 ($K=4$) and then stops improving (0.40 at $K=6$), matching the law with $v_K = \text{Var}(u | X, \hat{\gamma}_K)$ (0.78, 0.49, 0.37, 0.38). For a finite-state chain, $v = \mathbb{E}\text{Var}(S | Z_{1:N})$ is the minimum mean-squared smoothing error, so it is at most the linear Wiener error of the rescaled indicator: $v \leq \frac{1}{4} v_W(2\rho - 1, \iota)$ with $\iota = \Delta_Z^2 h^\top R^{-1}h$ for emission means $\pm \Delta_Z h$ (a bound, valid for known parameters). It is tight for weak signals (ratio 1.04–1.4 for $\Delta_Z \leq 0.5$, $\rho = 0.9$) and loose for well-separated regimes, where v falls exponentially (exact $v = 10^{-4}$ vs. bound 0.017 at $\Delta_Z = 2$; App. C). Persistent discrete regimes with informative proxies are therefore far less dangerous than a continuous latent state with the same sensor information, whose bias decays only polynomially.

Table 1: Bias predicted from primitives vs. observed ($\Delta_g=3$, $\Delta_T=1$, $\sigma^2=1$).

ϕ	ι	closed-form v	predicted bias	observed bias
0.9	0.69	0.261	0.622	0.601
0.9	2.76	0.128	0.340	0.337
0.9	11.04	0.055	0.156	0.154
0.97	0.69	0.146	0.383	0.373
0.97	2.76	0.073	0.203	0.219
0.97	11.04	0.034	0.100	0.107

6 BEYOND SENSITIVITY: WHEN CAN b BE LEARNED?

The law leaves b unidentified. Two routes learn it, and they fail differently. **Joint model.** Treat (Z, T, Y) as emissions of the chain (a Markov-switching regression with a common θ) and maximize the likelihood by EM. When the regime model is correct this is essentially unbiased and has the smallest variance in our simulations, even when Z is uninformative (App. C); the price of the modular, two-stage posterior used above is exactly the law. But it inherits the regime model: if the latent confounder is a continuous AR(1) process, a $K=2$ joint fit is as biased as posterior-DML (about +0.8 on $\theta=1$). **Proximal route.** Suppose a negative-control outcome $W = cS + \ell T + e$ is available ($c \neq 0$; $\ell = 0$ means T does not cause W), and $Z \perp (U, V, e) \mid S$.

Proposition 4 (Proximal identification, linear case). *Under the linear model of Sec. 2, the 2SLS regression of $\tilde{Y}$ on $(\tilde{T}, \tilde{W})$ with instruments $(\tilde{T}, \tilde{\gamma})$, where tildes denote cross-fitted residuals on X , converges to $(\theta - \ell b/c, b/c)$. Hence $\ell = 0$ identifies θ and b without any model for the latent state, and for $\ell \neq 0$ the bias is $-\ell \hat{\beta}_W$, computable for any hypothesized leakage ℓ .*

This is the linear special case of proximal causal inference (Miao et al., 2018; Tchetgen Tchetgen et al., 2020); we claim no novelty for the identification argument. What the regime setting adds is (i) the posterior $\hat{\gamma}$ is a valid instrument when W is an external variable, but *not* when W is a lagged outcome, because a smoothed posterior uses past Z that also drives W 's error (we found a bias of about 0.2 that persists at $N = 40,000$; raw Z_t is valid), and (ii) in a continuous-latent stress test, proximal 2SLS has bias below 0.03 where posterior-DML and the joint MLE are off by 0.8–0.9 (Table 7). The law still describes the failure: replacing S by the continuous state u and v by $\text{Var}(u \mid X, \hat{\gamma})$ predicts the bias within 4% (0.75 vs. 0.77 and 0.89 vs. 0.90). The cost is variance (RMSE 0.02–0.16 vs. 0.01–0.03 for the correct joint model) and reliance on a valid negative control; a leaking control ($\ell = 0.3$) gives bias $-0.45 = -\ell b/c$, as predicted. We tried the lagged outcome Y_{t-h} as a self-contained negative control: it

Table 2: Agreement between measured bias and law($\hat{v}$).

Setting	cells	corr($\hat{\theta}-\theta$, law($\hat{v}$))	RMSE	med. rel. err.	corr (law(v_N))
$K=2$, filter, lin	24	0.997	0.016	7.3%	1.000
$K=2$, smooth, gbm	24	0.998	0.011	4.3%	0.995
$K=2$, smooth, lin	24	0.998	0.009	5.1%	0.999
$K=3$, smooth, lin	8	0.999	0.010	9.8%	0.998

works for iid shocks and small Δ_T but is not robust to autocorrelated shocks (bias 0.07–0.21), so we do not recommend it. On the four cities we do not apply the proximal estimator: the available proxies (meteorology) also affect $\text{PM}_{2.5}$ directly, so the exclusion restrictions fail.

7 EXPERIMENTS

Setup. All experiments use `src/bias_law/` and share these ingredients unless noted: a persistent Markov chain (stay probability 0.9), 2-d Gaussian proxies with separation Δ_Z , two exogenous controls, $N = 3000$, unit-variance noises, $\Delta_g = 3$. The HMM is fitted by EM on the same sample (smoother or filter), cross-fitting uses 5 purged blocks with $\tau = 24$, and nuisances are linear or gradient boosting. Reported biases are means over 12 replications per cell with $\theta = 1$; the law is evaluated with the *observable* plug-ins unless stated.

7.1 The law predicts the bias

Figure 1 and Table 2 summarize 72 cells for $K = 2$ (smoother and filter; linear and boosted nuisances; three proxy qualities $\Delta_Z \in \{0.5, 0.8, 1.2\}$; eight values of Δ_T from 0.25 to 32) and 8 cells for $K = 3$ (matrix law, four (a, b) configurations including sign changes). The law with the observable plug-in $\hat{v}$ has correlation 0.996 and RMSE 0.012 with the measured bias, with median relative error 5.2% on cells with $|\text{bias}| > 0.05$. Using the realized v_N gives 3.5%, and also estimating $\hat{a}$ and $\widehat{\text{Var}}(\tilde{T})$ from the data gives 14%. The predicted peak location $\sigma/\sqrt{\hat{v}}$ and height $|\Delta_g|\sqrt{\hat{v}}/2\sigma$ match the observed maximum to within the grid (Appendix D).

Soft beats hard. In all 24 (Δ_Z, Δ_T) cells the soft posterior has smaller $|\text{bias}|$ than the hard regime label (median ratio 1.19 where the bias exceeds 0.02), as Prop. 1 predicts; DML on X alone has mean absolute bias 0.39 versus 0.14 for the soft posterior and 0.01 for the oracle state (Fig. 1c).

Using the law to correct. Subtracting the plug-in bias $\hat{a} \hat{v} b / \widehat{\text{Var}}(\tilde{T})$ with the true b reduces the per-replication mean absolute error from 0.126 to 0.021 (83%). Misspecifying b by a factor of 0.5 or 2 gives 0.072 and 0.097 (43% and 23% below the uncorrected error). The correction therefore degrades gracefully but is not robust to gross misspecification, which is

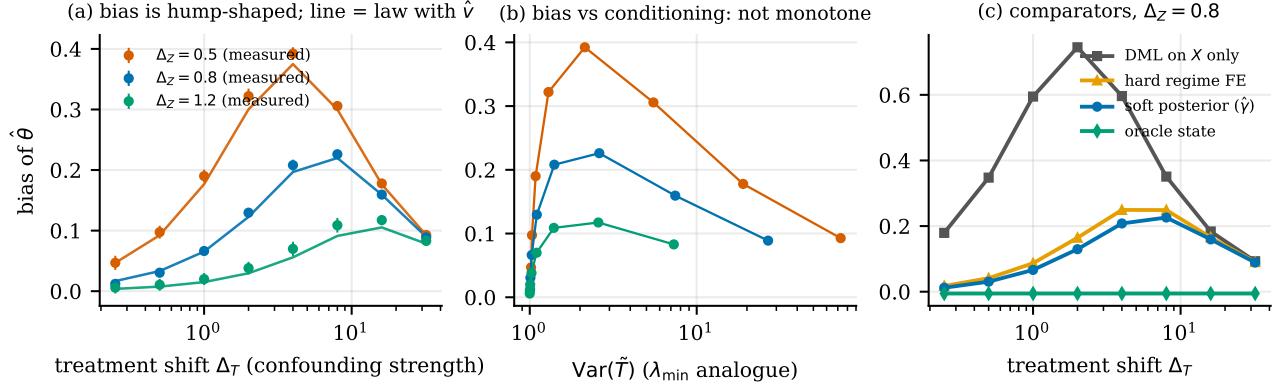


Figure 1: Bias of the soft-conditioning estimator against the treatment shift Δ_T ($K = 2$, learned smoother, linear nuisances, 12 replications, 95% bands). (a) The measured bias is hump-shaped and matches the law (lines) with $\hat{v}$ read from the posterior. (b) The same bias against $\text{Var}(\tilde{T})$, the scalar analogue of $\lambda_{\min}(\hat{J})$: conditioning rises monotonically while bias rises and then falls. (c) Hard regime fixed effects are worse than the soft posterior; DML on X alone is far worse; the oracle state is unbiased.

Table 3: Table-1 DGP (30 replications; $N = 1200$; law uses the realized v_N). θ estimates; “excess” = $\hat{\theta}_{\text{soft}} - \hat{\theta}_{\text{oracle}}$; true pooled $\hat{\theta} = 1.53$.

Δ_Z	DML on X	soft $\hat{\gamma}$	oracle S	excess bias	law(v_N)	law($\hat{v}$)
0.2	10.02	9.773	1.535	8.238	8.236	7.418
0.5	10.02	7.776	1.535	6.241	6.267	5.655
1.0	10.02	3.796	1.535	2.260	2.299	1.891
1.5	10.02	1.923	1.535	0.388	0.402	0.320
2.0	10.02	1.559	1.535	0.024	0.024	0.016
4.0	10.02	1.535	1.535	0.000	0.000	0.000

why we advocate sensitivity analysis rather than point correction.

7.2 It explains a standard difficulty benchmark

We rerun our earlier difficulty-frontier DGP ($\theta = (0.75, 2.5)$, treatment shift $\Delta_T = 4$, outcome offsets 15/50, i.e. $\Delta_g = 35$), in which “DML on X only” has bias ≈ 8.5 . Table 3 shows that the *excess* bias of soft conditioning over the oracle-state estimator is predicted by the heterogeneous-effects form of the law (with the realized v_N) to within 0.04 at every Δ_Z , from 8.24 at $\Delta_Z = 0.2$ to 0.02 at $\Delta_Z = 2$; the observable plug-in $\hat{v}$ underestimates it by 10–20%. So the transition between failure and success that this benchmark was built to display is a consequence of $v(\Delta_Z)$, not of a spectral property of the Jacobian.

Honest baselines. With the proxy used as controls, a regime-fixed-effects baseline matches the coupled OR-DML estimator at $\Delta_Z \geq 2$ (median absolute bias 0.063 vs. 0.072 at $\Delta_Z = 2$, 0.046 vs. 0.046 at $\Delta_Z = 4$) and

Table 4: Task-conditioning sweep. Regime-weighted score on the Table-1 DGP, $\Delta_Z = 1.5$.

Δ_T	nuisances on X only $\lambda_{\min}(\hat{J})$	$\ \hat{\theta} - \theta\ _2$	nuisances on (X, S) $\lambda_{\min}(\hat{J})$	$\ \hat{\theta} - \theta\ _2$
0.4	0.54	5.92	0.51	0.10
0.8	0.61	10.34	0.51	0.08
1.5	0.86	14.05	0.51	0.07
3.0	1.88	13.38	0.51	0.07
6.0	5.61	8.76	0.51	0.07
10.0	14.38	5.73	0.51	0.06

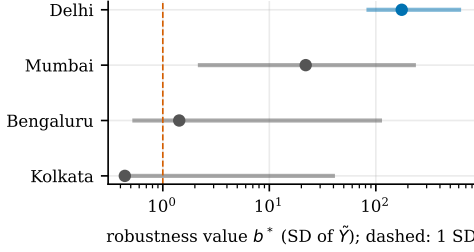
beats it at $\Delta_Z = 1$ (2.64 vs. 4.02); the spectral penalty changes OR-DML by less than 1.5% (Appendix E). The contribution of this paper is therefore the law and the sensitivity analysis, not a new estimator.

7.3 Why $\lambda_{\min}$ misleads

A common experiment varies the treatment shift while holding the representation fixed and reads the error against $\lambda_{\min}(\hat{J})$. Table 4 (12 replications per cell, boosted nuisances, oracle regime weights) shows the pitfall: if nuisances are fit on X only, the regime shift leaks into $\tilde{T}$, $\lambda_{\min}$ grows from 0.54 to 14.4, and the error is hump-shaped (5.9, 14.0, 5.7). If the nuisances are fit on (X, S) , $\lambda_{\min}$ is flat at 0.51 and the error is 0.06–0.10 for every Δ_T . The hump is the one of Cor. 1 with $v = \text{Var}(S)$, whose predicted peak $\sigma/\sqrt{v} \approx 2$ matches the observed peak at 1.5–3. Because $\lambda_{\min}$ moves while the error is flat or falls, $\lambda_{\min}$ does not govern reliability here.

Table 5: Temperature scaling ($K = 2$, boosted nuisances, 20 replications).

τ	bias	law($\hat{v}$)	law(v_N)	$\hat{v}$	v_N	$\sqrt{CE} + CE$
0.30	0.128	0.033	0.160	0.006	0.030	0.071
0.50	0.127	0.056	0.160	0.010	0.030	0.053
0.75	0.127	0.088	0.160	0.016	0.030	0.035
1.00	0.127	0.123	0.160	0.022	0.030	0.025
1.50	0.127	0.201	0.160	0.039	0.030	0.038
2.50	0.127	0.355	0.160	0.077	0.030	0.098


Figure 2: Robustness values b^* (log scale) with 90% block-bootstrap intervals; blue: $\hat{\theta}$ clearly nonzero.

7.4 Calibration controls the observable bias

We rescale the posterior logits by a temperature τ ($\tau < 1$ over-confident) and compare the law with $\hat{v}$ against the realized residual variance (Table 5). A flexible learner is invariant to monotone recalibration, so the realized bias is constant (0.127) while $\hat{v}$ changes by a factor of 14; the bound $|\hat{v} - v| \leq \sqrt{CE} + CE$ held in 100% of the 120 runs. An over-confident posterior therefore understates the bias, and the sensitivity analysis should use a recalibrated posterior.

7.5 Four Indian cities

We apply the analysis to hourly NO_2 (treatment) and $\text{PM}_{2.5}$ (outcome) in Delhi, Mumbai, Bengaluru and Kolkata (Feb 2025–Jun 2026) with the controls and proxies of Appendix F. Figure 2 and Table 6 report the pooled estimate and the robustness value. For Delhi the NO_2 – $\text{PM}_{2.5}$ association is positive and stable, with a robustness value of about 175 standard deviations of $\tilde{Y}$ (90% interval 85–605), about 1 200 times the observed regime shift of $\text{PM}_{2.5}$ ($\hat{c}_Y$, last column): residual confounding through the regime channel cannot plausibly overturn it. Mumbai’s and Kolkata’s intervals include zero, and Bengaluru’s estimate is ≈ 0 , so their robustness values are uninformative. Screening runs of ≥ 6 identical NO_2 readings changes Mumbai’s estimate from 1.9 to 4.9 (885 rows, 21%), which shows that the estimate is sensitive to data quality; Kolkata’s NO_2 has 31 distinct values and was nearly constant.

Table 6: Real-data sensitivity (pooled soft-conditioning estimate, Ridge nuisances); full table in Appendix F.

City	screened	$\hat{\theta}$ [90% CI]	b^* (SD) [90% CI]	$b^*/ \hat{c}_Y $
Delhi	no	0.58 [0.45, 0.68]	175.0 [85.4, 605]	1228
Mumbai	no	1.90 [-1.67, 5.01]	22.0 [2.2, 227]	366
Bengaluru	no	-0.01 [-0.15, 0.14]	1.4 [0.5, 109]	5
Kolkata	no	0.12 [-2.57, 3.33]	0.4 [0.5, 40]	1
Mumbai	yes	4.89 [-0.12, 13.74]	31.0 [2.8, 99]	417
Bengaluru	yes	-0.00 [-0.17, 0.13]	0.2 [2.2, 357]	1
Kolkata	yes	0.41 [-2.70, 3.52]	1.5 [0.5, 51]	3

8 DISCUSSION AND LIMITATIONS

We gave an exact, observable-except-one-parameter law for the bias of soft conditioning on a latent Markov posterior. It explains when posterior-based DML fails, why a standard conditioning diagnostic is not monotone in reliability, and how to bound the damage. **Practical guidance.** (a) Report $\hat{v}$, $\hat{a}$ and the robustness value alongside every posterior-conditioned effect. (b) Do not read $\lambda_{\min}$ of the residualized Gram matrix as a reliability index when a latent confounder may remain; fit nuisances on $(X, \hat{\gamma})$. (c) Prefer soft posteriors to hard labels, recalibrate over-confident posteriors, and always compare with regime-fixed-effects and proxy-as-control baselines. **Limitations.** (1) The law is a probability-limit identity under Assumptions 1–3; it is not a finite-sample or uniform-over-HMM-error guarantee, and Assumption 2 (locality of the posterior) is asserted via filter stability, not proved for EM-fitted parameters. (2) Assumption 1 (treatment noise independent of the proxies) is strong; correlation between the NO_2 innovations and meteorology would add terms. (3) The model is linear in the regime indicators and has a single scalar treatment. (4) b is unidentified, so the sensitivity analysis bounds rather than corrects. (5) The plug-in $\hat{v}$ requires calibration and is defined relative to the nuisance learner class: in a test where X_1 informs the regime, the Z -only $\hat{v}$ predicted the bias within 3% (0.30 vs. 0.31), whereas an HMM refit on (Z, X) gave a smaller $\hat{v}$ and underpredicted it (0.24), because a linear nuisance cannot exploit the information the HMM uses. (6) The real-data analysis is illustrative: the proxies partly overlap the controls, lagged pollutants may be mediators, and data-quality issues are material. (7) The formula is an OVB identity; the new content is its calibrated, dependent-data form and its consequences for DML diagnostics.

AI Use Statement

We used large language models to assist with code, derivations, and drafting. The authors reviewed the mathematical arguments, ran all experiments, and verified every reported number against the files in `reports/` and `reports/bias_law/`. The authors take full respon-

sibility for the content.

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CHECKLIST

1. For all models and algorithms presented, check if you include:
 - (a) A clear description of the algorithm, mapping, or model. **[Yes]** Section 2, Section 4.
 - (b) An analysis of the properties and complexity. **[Yes]** Theorem 1, Corollary 1, Propositions 1–2.
 - (c) (Optional) Anonymized source code, with specification of all dependencies, including external libraries. **[Yes]** Supplementary material (`src/bias_law/`, `requirements.txt`).
2. For any theoretical claim, check if you include:
 - (a) Statements of the full set of assumptions of all theoretical results. **[Yes]** Assumptions 1–3; strong ones are listed in the limitations.
 - (b) Complete proofs of all theoretical results. **[Partial]** Appendix A gives the algebra in full; the mixing/cross-fitting steps follow standard arguments and are given in outline.
 - (c) Clear explanations of any assumptions. **[Yes]** Section 3 and Section 6.
3. For all figures and tables that present empirical results, check if you include:
 - (a) The code, data, and instructions needed to reproduce the main experimental results. **[Yes]** `README.md`; `python src/bias_law/run_experiments.py`.
 - (b) All the training details (e.g., data splits, hyperparameters, how they were selected). **[Yes]** Section 7, Appendix D.
 - (c) A clear definition of the specific measure or statistics and error bars (e.g., confidence intervals, p-values). **[Yes]** Replication counts and bands are stated; real-data intervals are block bootstrap.
 - (d) A description of the computing infrastructure used. **[Yes]** Appendix D (2-CPU container, a few minutes per experiment).
4. If you are using existing assets (e.g., code, data, models) or curating/releasing new assets:
 - (a) If your work uses existing assets, did you cite the creators? **[Yes]** Data sources are described in Appendix F.
 - (b) Did you mention the license of the assets? **[Not Applicable]** Public environmental measurements.
 - (c) Did you include any new assets either in the supplemental material or as a URL? **[Yes]** Cleaned data and code in the supplement.
- (d) Did you discuss whether and how consent was obtained from people whose data you’re using/curating? **[Not Applicable]**
 - (e) Did you discuss whether the data you are using/curating contains personally identifiable information or offensive content? **[Not Applicable]**
5. If you used crowdsourcing or conducted research with human subjects: **[Not Applicable]**

A PROOFS

Proof of Theorem 1. *Step 0 (exact decomposition).* By Assumption 1, V_t is independent of $\mathcal{G} = \sigma(S, Z, X)$, which contains F_t and D_t , so $\mathbb{E}[T_t | F_t] = m(X_t) + a^\top c_t$ and $\mathbb{E}[Y_t | F_t] = \theta \mathbb{E}[T_t | F_t] + g(X_t) + b^\top c_t$. With nuisance errors $\delta_t^T = \hat{m}_T(F_t) - \mathbb{E}[T_t | F_t]$, $\delta_t^Y = \hat{m}_Y(F_t) - \mathbb{E}[Y_t | F_t]$ and $\delta^h = \delta^Y - \theta \delta^T$,

$$\tilde{T}_t = a^\top \tilde{D}_t + V_t - \delta_t^T, \quad \tilde{Y}_t - \theta \tilde{T}_t = b^\top \tilde{D}_t + U_t - \delta_t^h,$$

and $\hat{\theta} - \theta = \sum_t \tilde{T}_t (\tilde{Y}_t - \theta \tilde{T}_t) / \sum_t \tilde{T}_t^2$ exactly.

Step 1 (main terms). $N^{-1} \sum (a^\top \tilde{D}_t)(b^\top \tilde{D}_t) = a^\top \Sigma_N b$ and $N^{-1} \sum (a^\top \tilde{D}_t)^2 = a^\top \Sigma_N a$. Also $N^{-1} \sum V_t^2 = \sigma^2 + O_P(N^{-1/2})$ by the summable-autocovariance condition.

Step 2 (noise cross terms). Conditionally on $\mathcal{G}$, $\sum_t V_t b^\top \tilde{D}_t$ and $\sum_t a^\top \tilde{D}_t U_t$ have mean zero (Assumption 1) and variance at most $C \sum_t \|\tilde{D}_t\|^2 \leq CN$ by summable autocovariances; similarly $\sum_t V_t U_t$ (conditioning on $\mathcal{G}$ and V). Hence each is $O_P(N^{1/2})$.

Step 3 (nuisance terms). Within a cross-fitting fold, $\delta_t = \delta(F_t)$ is a fixed function given the training folds, and $\mathbb{E}[\tilde{D}_t | F_t] = 0$. The sums $\sum \delta_t \tilde{D}_t$, $\sum \delta_t V_t$, $\sum \delta_t U_t$ have conditional mean zero up to an $O(\varepsilon_\tau)$ dependence between training and test blocks (embargo plus Assumption 2) and variance $\leq CN \|\delta\|_{L^2}^2$ by the mixing covariance inequality (Davydov), so they are $O_P(rN^{1/2})$. The product term satisfies $|\sum \delta_t^T \delta_t^h| \leq N r_T r_h$ by Cauchy-Schwarz.

Step 4. Write the ratio as $(A + e_1)/(B + e_2)$ with $A = a^\top \Sigma_N b$, $B = a^\top \Sigma_N a + \sigma^2 \geq \sigma^2$, and $e_1, e_2 = O_P(N^{-1/2} + r_T r_h + \varepsilon_\tau)$. Then $(A + e_1)/(B + e_2) - A/B = (e_1 - (A/B)e_2)/(B + e_2)$ and $|A/B| \leq \sqrt{b^\top \Sigma_N b}/(2\sigma)$ is bounded (Corollary 1(i), matrix form via Cauchy-Schwarz and $x/(x^2 + \sigma^2) \leq 1/(2\sigma)$). $\square$

Proof of Corollary 1. (i) $a^2 v + \sigma^2 \geq 2|a|\sqrt{v}\sigma$ (AM-GM), with equality at $|a| = \sigma/\sqrt{v}$. (ii) $|a| = \sqrt{(\lambda - \sigma^2)/v}$ gives $|B| = |b|v|a|/\lambda$; and $\frac{d}{d\lambda} \sqrt{\lambda - \sigma^2}/\lambda = (2\sigma^2 - \lambda)/(2\lambda^2 \sqrt{\lambda - \sigma^2})$. (iii) $\partial_v [v/(a^2 v + \sigma^2)] = \sigma^2/(a^2 v + \sigma^2)^2 > 0$. $\square$

Proof of Proposition 1. For $\mathcal{F} \subseteq \mathcal{F}'$ the law of total variance gives $\mathbb{E}\text{Cov}(D | \mathcal{F}') \preceq \mathbb{E}\text{Cov}(D | \mathcal{F})$; hard labels are a measurable function of $\hat{\gamma}$, hence $\sigma(\text{hard}) \subseteq \sigma(\hat{\gamma})$. Part (iii) of Corollary 1 finishes the $K = 2$ case. $\square$

Proof of Proposition 2. With $c = \mathbb{E}[D | F]$, $\gamma(1 - \gamma) - c(1 - c) = (\gamma - c)(1 - 2c) - (\gamma - c)^2$, so $|\mathbb{E}[\gamma(1 - \gamma) - c(1 - c)]| \leq \mathbb{E}|\gamma - c| + \mathbb{E}(\gamma - c)^2 \leq \sqrt{CE} + CE$ (Cauchy-Schwarz, $|1 - 2c| \leq 1$), and $\mathbb{E}c(1 - c) = \mathbb{E}\text{Var}(D | F) = v$. If m is linear and $c = \hat{\gamma}$, then $\mathbb{E}[T | F] = m(X) + a^\top \hat{\gamma}$ is linear in $(X, \hat{\gamma})$ and the OLS coefficient is a ; $\text{Var}(\tilde{T}) = a^\top \Sigma a + \sigma^2$ is Step 1. $\square$

B HETEROGENEOUS EFFECTS AND MULTIPLE REGIMES

Heterogeneous effects ($K = 2$). Let $Y_t = \theta(S_t)T_t + g(X_t) + bD_t + U_t$ with $\theta(0) = \theta_0, \theta(1) = \theta_1, \bar{\theta} = \mathbb{E}\theta(S)$, $T_t = \bar{m} + m_0(X_t) + aD_t + V_t$ with $\mathbb{E}m_0 = 0$, and assume $\mathbb{E}[m_0(X)\text{Var}(D | F)] = 0$. Then

$$\hat{\theta} \xrightarrow{p} \bar{\theta} + \frac{av[b + a(\theta_1 - \bar{\theta}) + \bar{m}(\theta_1 - \theta_0)]}{a^2 v + \sigma^2}.$$

Proof. $\mathbb{E}[\tilde{T}\tilde{Y}] = \mathbb{E}[\tilde{T}\theta(S)T] + av$; expanding with $\tilde{T} = a\tilde{D} + V$ gives $\mathbb{E}[\tilde{T}\theta(S)T] = a\bar{m}(\theta_1 - \theta_0)v + a^2\theta_1 v + \sigma^2\bar{\theta}$, using $\mathbb{E}[\tilde{D}\theta(S)] = (\theta_1 - \theta_0)v$ and $\mathbb{E}[V^2\theta(S)] = \sigma^2\bar{\theta}$. Subtract $\bar{\theta}(a^2 v + \sigma^2)$. $\square$ The oracle-state estimator ($v = 0$) converges to $\bar{\theta}$.

$K > 2$. Theorem 1 holds with matrix Σ_N and $|B| \leq \sqrt{b^\top \Sigma b}/(2\sigma)$; the hump persists along any direction $a = s a_0$. Hard-versus-soft monotonicity (Prop. 1) holds in the Loewner order but $|B|$ need not be monotone in Σ for $K > 2$.

C EXTENDED COMPARISON OF ESTIMATORS

Bias (RMSE) of $\hat{\theta} - \theta$ with $\theta=1$ (ATE target for the heterogeneous row), $N=3000$, 12 replications ($\Delta_g=3$, $\Delta_Z=0.5$, persistence 0.9; continuous rows: 12 replications, AR(1) coefficient 0.97). Posterior-DML uses cross-fitted

nuisances on $(X, \hat{\gamma})$. Code: `src/bias_law/ext*.py`; raw output `reports/bias_law/x[1-5]*.csv`. With Z uninformative ($\Delta_Z=0$) and $\Delta_g=3$ the joint MLE is still unbiased for every $\Delta_T \in \{0.5, \dots, 4\}$ (bias $|\cdot| \leq 0.02$) because the mixture in (T, Y) is identifiable under Gaussianity; non-identification without proxies therefore holds only without parametric assumptions on the mixture.

Table 7: Estimator comparison under misspecification.

Scenario	Δ_T	Posterior-DML	Joint MLE	Proximal 2SLS
HMM correctly specified	1	+0.20 (0.20)	+0.00 (0.02)	-0.01 (0.02)
HMM correctly specified	4	+0.40 (0.40)	+0.00 (0.02)	+0.02 (0.06)
nonlinear $f(X)$ (GBM nuisances)	1	+0.16 (0.17)	-0.01 (0.01)	-0.00 (0.04)
nonlinear $f(X)$ (GBM nuisances)	4	+0.37 (0.37)	-0.00 (0.01)	-0.03 (0.06)
heavy-tailed (t_3) proxy noise	1	+0.15 (0.16)	+0.01 (0.02)	+0.00 (0.04)
heavy-tailed (t_3) proxy noise	4	+0.32 (0.32)	-0.00 (0.01)	-0.04 (0.05)
true $K=3$, fitted $K=2$	1	+0.46 (0.47)	-0.00 (0.02)	-0.01 (0.05)
true $K=3$, fitted $K=2$	4	+0.56 (0.57)	+0.00 (0.01)	+0.06 (0.16)
heterogeneous $\theta(S)$ (ATE target)	1	+0.26 (0.26)	-0.01 (0.03)	-0.02 (0.06)
heterogeneous $\theta(S)$ (ATE target)	4	+0.66 (0.66)	-0.00 (0.02)	-0.01 (0.08)
negative control leaks ($W \ni 0.3T$)	1	+0.21 (0.21)	+0.01 (0.03)	-0.45 (0.45)
negative control leaks ($W \ni 0.3T$)	4	+0.39 (0.39)	-0.00 (0.01)	-0.46 (0.47)
continuous AR(1) latent, SNR 0.5	1	+0.83 (0.84)	+0.79 (0.80)	-0.02 (0.04)
continuous AR(1) latent, SNR 1	1	+0.77 (0.78)	+0.79 (0.79)	-0.02 (0.04)
continuous AR(1) latent, SNR 0.5	2	+0.90 (0.90)	+0.89 (0.89)	-0.03 (0.08)
continuous AR(1) latent, SNR 1	2	+0.87 (0.88)	+0.88 (0.89)	-0.02 (0.06)

D EXPERIMENTAL DETAILS

Learned-HMM grid. $K = 2$: $\Delta_Z \in \{0.5, 0.8, 1.2\}$, $\Delta_T \in \{0.25, 0.5, 1, 2, 4, 8, 16, 32\}$, 12 replications per cell, $N = 3000$, persistence 0.9, $\Delta_g = 3$, $\sigma = 1$. The HMM is fitted once per replication (Gaussian emissions, diagonal covariance, EM); smoothing and filtering posteriors are both used. Predicted vs. observed peak (value of Δ_T at the maximum measured bias, grid-limited): $\Delta_Z = 0.5$: 4.0 vs. 4.0; $\Delta_Z = 0.8$: 6.7 vs. 8; $\Delta_Z = 1.2$: 14.1 vs. 16 (linear nuisances, smoother); predicted peak heights 0.375/0.224/0.106 vs. observed 0.392/0.226/0.117. `reports/bias_law/e1_peak_check.csv`.

$K = 3$. Proxy means on a circle of radius $1.8\Delta_Z$; $(a, b) \in \{(0, 2, 5; 0, 3, 1), (0, 3, -3; 0, 2, 2), (0, 1, 1; 0, 3, -3), (0, 6, 3; 0, 3, 3)\}$; 20 replications; linear nuisances.

Table-1 DGP. `generate_difficulty_dgp` with $\Delta_T = 4$, outcome offsets 15 and 50, state-free treatment level $\bar{m} = 2$, $\theta = (0.75, 2.5)$; the oracle-state estimator provides $\hat{\theta}$.

Compute. 2-CPU container, single-threaded nuisances; each script runs in a few minutes. Seeds are fixed in the scripts.

E COUPLED OR-DML AND THE BENCHMARK

The coupled regime-weighted estimator $\hat{\theta} = (\hat{J} + \lambda I)^{-1} \hat{S}$ of our earlier work, with a spectral penalty $\lambda \propto N^{-1/2}$, is evaluated against honest baselines (Table 8; 100 replications per Δ_Z). At $\Delta_Z \geq 2$ it ties regime fixed effects built from the same HMM; at $\Delta_Z \leq 1$ every method fails, with regime fixed effects better at $\Delta_Z = 1$. The spectral penalty is numerically inert (the “spectral” and “unregularized” rows differ by ≤ 0.003 in mean absolute bias at $\Delta_Z = 2$). Coverage for the regime-FE row uses the blocked-DML standard error and should be read as indicative only. We therefore do not present the coupled estimator as a contribution.

Table 8: Difficulty-frontier benchmark (100 replications per Δ_Z , $N = 1200$, true $\theta_{\text{ATE}} = 1.5278$); `reports/or_dml_benchmark_summary.csv`.

Δ_Z	Estimator	mean bias	median bias	RMSE	cover. SATE
0.2	Oracle DML	0.056	0.044	0.070	96%
	Standard DML	8.456	8.456	8.456	0%
	Block DML	8.490	8.479	8.491	0%
	DML + Z Controls (Ridge)	8.307	8.312	8.308	0%
	DML + Z Controls (GBM)	8.018	8.025	8.019	0%
	Regime FE DML (Hard)	8.277	8.281	8.278	0%
	Unregularized Coupled DML	8.640	8.613	8.641	0%
	Spectral OR-DML (Ours)	8.630	8.602	8.631	0%
	Filtered OR-DML (Ours)	8.606	8.587	8.607	0%
0.5	Oracle DML	0.055	0.044	0.071	92%
	Standard DML	8.429	8.432	8.430	0%
	Block DML	8.467	8.483	8.468	0%
	DML + Z Controls (Ridge)	7.429	7.424	7.430	0%
	DML + Z Controls (GBM)	7.031	7.020	7.034	0%
	Regime FE DML (Hard)	6.593	6.632	6.600	0%
	Unregularized Coupled DML	8.847	8.886	8.851	0%
	Spectral OR-DML (Ours)	8.835	8.874	8.839	0%
	Filtered OR-DML (Ours)	8.988	9.001	8.991	0%
1.0	Oracle DML	0.057	0.043	0.073	92%
	Standard DML	8.433	8.449	8.434	0%
	Block DML	8.466	8.475	8.467	0%
	DML + Z Controls (Ridge)	5.438	5.462	5.440	0%
	DML + Z Controls (GBM)	3.633	3.642	3.647	0%
	Regime FE DML (Hard)	2.642	2.689	2.679	0%
	Unregularized Coupled DML	4.025	4.030	4.045	0%
	Spectral OR-DML (Ours)	4.017	4.022	4.037	0%
	Filtered OR-DML (Ours)	5.172	5.185	5.185	0%
2.0	Oracle DML	0.052	0.042	0.066	93%
	Standard DML	8.432	8.451	8.433	0%
	Block DML	8.461	8.472	8.462	0%
	DML + Z Controls (Ridge)	2.631	2.614	2.635	0%
	DML + Z Controls (GBM)	0.207	0.194	0.243	72%
	Regime FE DML (Hard)	0.077	0.063	0.102	96%
	Unregularized Coupled DML	0.082	0.073	0.105	93%
	Spectral OR-DML (Ours)	0.081	0.072	0.104	92%
	Filtered OR-DML (Ours)	0.102	0.081	0.131	92%
4.0	Oracle DML	0.054	0.048	0.067	96%
	Standard DML	8.451	8.461	8.452	0%
	Block DML	8.486	8.495	8.487	0%
	DML + Z Controls (Ridge)	0.848	0.842	0.853	0%
	DML + Z Controls (GBM)	0.101	0.082	0.125	89%
	Regime FE DML (Hard)	0.054	0.046	0.067	100%
	Unregularized Coupled DML	0.054	0.048	0.067	96%
	Spectral OR-DML (Ours)	0.054	0.046	0.067	96%
	Filtered OR-DML (Ours)	0.054	0.046	0.067	96%

F REAL-DATA DETAILS

Data. 14 122 hourly records (CPCB monitors via OpenAQ; Open-Meteo reanalysis) for Delhi, Mumbai, Bengaluru, Kolkata. Treatment: NO_2 ; outcome: $\text{PM}_{2.5}$; proxies Z : temperature, wind speed (km/h), humidity, pressure, hour-of-day sine/cosine; controls X : lagged and rolling pollutants plus the meteorological variables (so Z overlaps X ; the plug-in $\hat{v}$ ignores the information in X and can overstate v). Sensor screening: runs of ≥ 6 identical NO_2 readings are *not* removed in the main analysis; Table 6 reports both. Delhi has none; Mumbai has a 273-hour constant run; Kolkata’s NO_2 has 31 distinct values. **Newey–West sensitivity of the regime-specific OR-DML estimates.** Standard errors are *not* stable across lags: for example the Bengaluru regime-2 effect has $p = 0.027$ at lag 12, 0.071 at lag 72 and 0.101 at lag 168; Mumbai’s regime-2 effect has $p = 0.026, 0.051, 0.074$. The regime-specific claims of our earlier work should be read accordingly.

Table 9: Full real-data sensitivity table; `reports/bias_low/e5_real_city_sensitivity.csv`.

City	screen	N	$\hat{\theta}$ [90% CI]	$\hat{v}$	$\hat{a}$	$\text{Var}(\bar{T})$	b^* (SD of $\bar{Y}$) [90% CI]	$b^*/ \hat{c}_Y $
Delhi	no	4640	0.584 [0.449, 0.682]	0.005	-4.399	278.54	175.02 [85.37, 604.64]	1228
Mumbai	no	4180	1.903 [-1.672, 5.014]	0.035	-0.033	0.69	21.96 [2.23, 227.45]	366
Bengaluru	no	4093	-0.006 [-0.148, 0.135]	0.025	-0.355	12.21	1.43 [0.54, 109.33]	5
Kolkata	no	1209	0.117 [-2.572, 3.329]	0.026	0.029	0.01	0.44 [0.47, 39.51]	1
Delhi	yes	4640	0.584 [0.449, 0.682]	0.005	-4.399	278.54	175.02 [85.37, 604.64]	1228
Mumbai	yes	3295	4.893 [-0.123, 13.744]	0.017	0.130	0.74	31.01 [2.76, 98.92]	417
Bengaluru	yes	3930	-0.000 [-0.174, 0.128]	0.024	-0.203	12.45	0.20 [2.18, 357.38]	1
Kolkata	yes	1099	0.415 [-2.704, 3.516]	0.024	0.034	0.01	1.49 [0.46, 51.03]	3

Table 10: HAC-lag sensitivity (hours); `reports/bias_low/e6_hac_lag_sensitivity.csv`.

City	regime	$\hat{\theta}$	SE (12)	SE (24)	SE (72)	SE (168)
Delhi	1	0.704	0.051	0.055	0.063	0.073
Delhi	2	0.128	0.074	0.083	0.109	0.125
Mumbai	1	-0.030	8.716	6.437	4.076	3.053
Mumbai	2	3.303	1.480	1.538	1.692	1.847
Bengaluru	1	0.092	0.155	0.167	0.182	0.167
Bengaluru	2	-0.153	0.069	0.072	0.084	0.093
Kolkata	1	0.679	1.404	1.344	1.338	1.201
Kolkata	2	-1.065	2.668	2.757	3.082	3.346